

A Strict Endpoint-Budget Compiler for the Literal V59 Four-Packet Interface

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August 26, 2026

Abstract

The recent V59 analysis has reached a precise interface obstruction. A four-packet DFT identifies mode zero as the missing datum, while the known same-clock null suppression is only logarithmic. This paper compiles the endpoint arithmetic into an exact finite-lane budget. The current baseline is $E_0 = 5/3$ and the target is $E_* = 1997/1200$, so the required strict saving is $E_0 - E_* = 1/400$. If lane j saves δ_j powers and pays a reassembly loss λ_j , the effective credit is $\sigma_j = \delta_j - \lambda_j$ and a finite reassembly closes the target only when $\min_j \sigma_j > 1/400$. Equality is borderline at the power level. We also prove that any fixed logarithmic saving has zero fixed-power credit. Finally, a scaled version of the exact TPC-260 plus/alternating witness has identical packet marginals and Haar/null data but full energies $16x^{5/3}$ and 0. This is a structural synthetic obstruction, not a prime-shell counterexample: the literal mode-zero or signed cross-Gram estimate remains open.

1 Scope and motivation

The Route-B object is a signed coupling built on one fixed V59 clock. The four-block Haar construction of TPC-257 and the source-frozen transverse null direction of TPC-258–259 isolate a useful channel, but they do not by themselves control the whole output. TPC-260 made the missing information explicit. For four packets, packet diagonal energy fixes the sum of Fourier mode energies, whereas full reassembly asks for mode zero; equal packet norms and all currently observed contrast data can coexist with full residual energy zero or sixteen.

The present paper answers the next, narrower question: exactly how much effective power saving would a new literal theorem have to provide after every lane loss? This question is elementary once stated, but it is operationally important. It prevents a logarithmic $o(1)$ statement, a local boundary gap, or a conditional comparison estimate from being silently counted as global endpoint credit.

Our contributions are:

1. an exact baseline-to-target exponent identity;
2. a strict finite-lane endpoint compiler with a sharp $1/400$ threshold;
3. a proof-level separation between logarithmic and fixed-power savings;
4. a scaled, null-compatible four-packet obstruction; and
5. a minimum-sufficient formulation of the next literal theorem.

All claims are deliberately scoped. No result here proves arithmetic L^2 , full Gate B, fixed-atom credit, or the twin-prime conjecture.

2 Endpoint normalization

We use the unnormalized exponent ledger inherited from the current big road:

$$E_0 = \frac{5}{3} = \frac{2000}{1200}, \quad E_* = \frac{1997}{1200}. \quad (1)$$

Subtracting gives the exact identity

$$\Delta_* := E_0 - E_* = \frac{3}{1200} = \frac{1}{400}. \quad (2)$$

The denominator 1200 is only a common bookkeeping denominator; no approximation is involved. The strict inequality in the next theorem is essential because the available analytic estimates are conventionally stated with an arbitrary x^ε loss.

Let $\mathcal{L}$ be a finite set of lanes. A lane may represent a packet, boundary, mask, deleted diagonal, or a signed reassembly component. The symbols δ_j and λ_j refer to proved quantities on the same clock; they are not labels for hoped-for estimates.

Definition 1 (effective lane credit). Suppose lane j has a saving δ_j and a loss λ_j . Its effective credit and the bottleneck credit are

$$\sigma_j = \delta_j - \lambda_j, \quad \sigma = \min_{j \in \mathcal{L}} \sigma_j. \quad (3)$$

3 The strict budget compiler

Theorem 1 (finite-lane endpoint compiler). *Let $\mathcal{L}$ be finite and nonempty. Suppose that for every $j \in \mathcal{L}$ and every $\varepsilon > 0$, there is a constant $C_{j,\varepsilon}$ such that, for all sufficiently large x ,*

$$|T_j(x)| \leq C_{j,\varepsilon} x^{E_0 - \delta_j + \lambda_j + \varepsilon}. \quad (4)$$

If $\sigma > \Delta_ = 1/400$, then*

$$\sum_{j \in \mathcal{L}} |T_j(x)| = o(x^{E_*}). \quad (5)$$

If $\sigma = \Delta_$, the hypotheses are only borderline at the power level; if $\sigma < \Delta_*$, the power ledger does not close.*

Proof. Put $\eta = \sigma - \Delta_* > 0$ and choose $\varepsilon = \eta/2$. Since $\sigma_j \geq \sigma$,

$$\begin{aligned} E_0 - \delta_j + \lambda_j + \varepsilon &= E_0 - \sigma_j + \varepsilon \\ &\leq E_0 - \sigma + \varepsilon = E_* - \eta/2. \end{aligned}$$

Thus every term in the finite sum is $O(x^{E_* - \eta/2})$, and the sum has the same bound. Dividing by x^{E_*} gives $O(x^{-\eta/2}) \rightarrow 0$. At equality, the displayed hypotheses leave exponent $E_* + \varepsilon$ for arbitrary $\varepsilon > 0$ but supply no negative fixed margin. Below equality the resulting exponent is larger than the target. $\square$

Table 1: Exact budget examples. “Local” and “conditional” labels do not transfer credit to the current literal whole object.

Lane	δ	λ	σ	classification
Conditional comparison	1/100	1/1200	11/1200	strict, conditional
Exact threshold	1/400	0	1/400	borderline
Threshold with loss	1/400	1/1200	1/600	insufficient
TPC-257 local boundary	1/48	0	1/48	strict, local only
TPC-259 log-only null	0	0	0	no fixed power

The theorem is a compiler, not an estimate of any particular lane. Its main use is to expose the exact obligation: every lane must be attached to one common literal object, and the smallest effective saving must be strictly larger than 1/400.

For the first row, the strict margin is $11/1200 - 1/400 = 1/150$. The local boundary row has the larger margin $1/48 - 1/400 = 11/600$, but it is not a global credit until a transfer theorem attaches that local estimate to the complete four-packet output.

4 The logarithmic firewall

Theorem 2 (logarithmic savings do not buy a power). *For every fixed $M > 0$ and every fixed $\delta > 0$,*

$$\frac{x^\delta}{(\log x)^M} \longrightarrow \infty. \quad (6)$$

Consequently a bound $x^{E_0}/(\log x)^M$ has zero fixed-power credit in the compiler of Theorem 1.

Proof. Taking logarithms of the left side of (6) gives $\delta \log x - M \log \log x$. Since $\log \log x / \log x \rightarrow 0$, this tends to infinity. If a logarithmic bound could pay a fixed δ power, one could choose an $\varepsilon < \delta$ and obtain a uniform comparison with $x^{E_0 - \delta + \varepsilon}$, contradicting (6). $\square$

This distinction is the relevant one for the current route. TPC-258 and TPC-259 provide a source-backed fixed-log suppression for a selected null channel. That is a genuine signed-coupling advance, but it records $\delta = 0$ in the fixed-power ledger. It cannot pay the strict 1/400 gap, and it does not estimate the orthogonal residual left by TPC-259.

5 A scaled four-packet obstruction

We now show that the endpoint firewall is not merely a bookkeeping convention. Let $z, w \in \mathcal{H}$ be orthonormal, with z the source-frozen null direction and w an orthogonal residual direction. In the four-block Haar space, w may be the normalized scaling mode. Set

$$a(x) = x^{5/6}, \quad V_j^+(x) = a(x)w, \quad V_j^-(x) = (-1)^j a(x)w, \quad 0 \leq j \leq 3. \quad (7)$$

Theorem 3 (scaled null-compatible witness). *The two families in (7) have the same packet diagonal, zero projection on every Haar contrast orthogonal to w , and zero projection on z . Their full residual energies are*

$$\left\| \sum_{j=0}^3 V_j^+(x) \right\|^2 = 16x^{5/3}, \quad \left\| \sum_{j=0}^3 V_j^-(x) \right\|^2 = 0. \quad (8)$$

Therefore those marginal and null data alone cannot imply a universal positive fixed-power saving for the full four-packet residual.

Proof. Each packet has norm $a(x)$, so each diagonal energy is $a(x)^2 = x^{5/3}$ in both families. Orthogonality gives all stated zero projections. The plus sum is $4a(x)w$, whereas the alternating sum cancels exactly. Squaring the norms gives (8). In particular, for any $\delta > 0$ the plus family is at baseline scale and is incompatible with a universal upper bound of order $x^{5/3-\delta}$ based only on these observables. $\square$

The witness is the TPC-260 finite construction with its amplitude scaled to the baseline exponent. It is a structural synthetic family, not a sequence of prime-shell outputs. The conclusion is therefore a scoped non-identifiability statement, not a proof that the literal route is impossible.

For completeness, the four-point DFT makes the same point in mode language:

$$\widehat{V}_k = \frac{1}{2} \sum_{j=0}^3 i^{-jk} V_j, \quad \sum_k \|\widehat{V}_k\|^2 = \sum_j \|V_j\|^2, \quad \left\| \sum_j V_j \right\|^2 = 4 \|\widehat{V}_0\|^2. \quad (9)$$

The plus family places all packet energy in mode zero, while the alternating family places it in mode two. The diagonal ledger sees the same total in both cases and cannot select the required mode.

6 Current route and the minimum sufficient theorem

The present evidence can be summarized as follows. The TPC-257 local boundary gap is $1/48$, the TPC-259 null channel is logarithmically small, and TPC-260 shows that the remaining four-packet mode zero is not determined by current marginals. These facts are compatible: a local gap is not a global saving, and null suppression is not residual reassembly.

The minimum sufficient next theorem is consequently precise:

Corollary 1 (minimum sufficient literal bridge). *A common-clock estimate for the literal V59 mode-zero output, or an equivalent signed cross-Gram estimate, with effective saving strictly greater than $1/400$ after all masks, deleted-diagonal, hard-window, boundary, and normalization losses, is sufficient to close the finite-lane endpoint compiler.*

The corollary is conditional on the displayed literal estimate; no such estimate is proved here. It also explains why the next route should not spend the same credit twice: the four packet lanes must be reassembled before a global endpoint claim is made.

Table 2: Claim ledger for TPC-261.

Item	Current status
Budget identity	proved exactly
Strict threshold	$\sigma > 1/400$, proved exactly
Log-only promotion	refuted in the scoped power sense
Scaled witness	exact structural synthetic certificate
Literal mode-zero estimate	open
Arithmetic L^2 / full Gate B	none / open
Twin-prime conclusion	none

7 Conclusion

TPC-261 turns the current endpoint intuition into a reusable theorem. The target gap is exactly $1/400$, and the correct currency is effective fixed-power credit, not an isolated logarithm or a local exponent. The scaled TPC-260 witness proves that the current marginal/Haar/null interface can leave a baseline-scale mode-zero residual. Thus the next genuine advance must estimate that literal mode or its signed cross-Gram on the common clock. Until then the arithmetic L^2 gate, full Gate B, fixed-atom credit, and the twin-prime conclusion remain open.

The release includes an exact rational certificate, an independent checker, and a stress audit. They verify the finite algebra and claim firewall only; no finite computation is treated as asymptotic evidence.

References

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